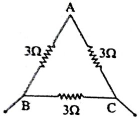
EXPLANATIONS

1. (*d*) The charge $q = \int I dt$.
With $I(t) = 0.02t + 0.01$ A,
integrate from t = 1s to t = 2s:
$$\int_0^2 (0.02t + 0.01)dt$$
$$q = \left[0.02\frac{t^2}{2} + 0.01t\right]_1^2$$
$= 0.01(3) + 0.01(1) = 0.04$ C

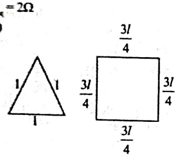
2. (*a*) Resistivity is independent of temperature for wire bound resistors.

3. [2]



two resistors (3Ω each) are in series, so their total resistance is:
$R_{total} = 3\Omega + 3\Omega = 6\Omega$
Now, the equivalent resistance between *B* and *C* is the parallel combination of the 3Ω and 6Ω resistances:
$$\frac{1}{R_{eq}} = \frac{1}{3\Omega} + \frac{1}{6\Omega}$$
Simplifying:
$R_{eq} = 2\Omega$

4. (*b*)



Let resistance corresponding to length l is R
Then for triangle
$$R_1 = \frac{\left(\frac{2R}{3}\right)\times\left(\frac{R}{3}\right)}{\left(\frac{3R}{3}\right)} = \boxed{\frac{2R}{9}}$$
For square
$$R_2 = \frac{3\left(\frac{R}{4}\right)\times\frac{R}{4}}{3\left(\frac{R}{4}\right)+\frac{R}{4}} = \frac{3}{16}R = \boxed{\frac{3R}{16}}$$
$$\frac{R_1}{R_2} = \left(\frac{\frac{2R}{9}}{\frac{3R}{16}}\right) = \frac{32}{27}$$

5. (*b*) Given that
Current $I = I_o + \beta t$
$I_o = 20$ A
$b = 3$A/s
$I = 20 + 3t$
$$\frac{dq}{dt} = 20 + 3t$$
$$\int_0^q dq = \int_0^{20} (20+3t)dt$$
$$q = \int_0^{20} 20dt + \int_0^{20} 3t\,dt$$
$$q = \left[20t + \frac{3t^2}{2}\right]_0^{20} = 1000C$$

6. (c) Given, $R_{T=27} = 60\ \Omega$
$V = 220$ V
$I_T = 2.75$ A
Hence,
$$R_T = \frac{220}{2.75} = 80\ \Omega$$
Now,
We know that,
$R = R_0(1 + \alpha\Delta T)$
Hence,
$80 = 60\,[1 + 2 \times 10^{-4}(T-27)]$
$T \approx 1694°$C

7. (*b*)For series combination,
$R_{eq} = R_1 + R_2$
$$2R(1+\alpha_{eq}\Delta\theta) = R(1+\alpha_1\Delta\theta) + R(1+\alpha_2\Delta\theta)$$
$$2R(1+\alpha_{eq}\Delta\theta) = 2R + (\alpha_1+\alpha_2)R\Delta\theta$$
$$\alpha_{eq} = \frac{\alpha_1+\alpha_2}{2}$$
For parallel combination,
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$
$$\frac{1}{\frac{R}{2}(1+\alpha_{eq}\Delta\theta)} = \frac{1}{R(1+\alpha_1\Delta\theta)} + \frac{1}{R(1+\alpha_2\Delta\theta)}$$
$$\frac{2}{1+\alpha_{eq}\Delta\theta} = \frac{1}{1+\alpha_1\Delta\theta} + \frac{1}{1+\alpha_2\Delta\theta}$$
$$\frac{2}{1+\alpha_{eq}\Delta\theta} = \frac{1+\alpha_2\Delta\theta+1+\alpha_1\Delta\theta}{(1+\alpha_1\Delta\theta)(1+\alpha_2\Delta\theta)}$$
$$2[(1+\alpha_1\Delta\theta)(1+\alpha_2\Delta\theta)]$$
$$= [2+(\alpha_1+\alpha_2)\Delta\theta][1+\alpha_{eq}\Delta\theta]$$
$$2[1+\alpha_1\Delta\theta+\alpha_2\Delta\theta+\alpha_1\alpha_2\Delta\theta]$$
$$=$$
$$2+2\alpha_{eq}\Delta\theta+(\alpha_1+\alpha_2)\Delta\theta + \alpha_{eq}(\alpha_1+\alpha_2)\Delta\theta^2$$
Neglecting small terms
$$2+2(\alpha_1+\alpha_2)\Delta\theta$$
$$= 2+2\alpha_{eq}\Delta\theta+(\alpha_1+\alpha_2)\Delta\theta$$
$$(\alpha_1+\alpha_2)\Delta\theta = 2\alpha_{eq}\Delta\theta$$
$$\alpha_{eq} = \frac{\alpha_1+\alpha_2}{2}$$

8. [22] $q = \int_1^2 i\,dt$
$$\Rightarrow q = \left(t^3+t^4\right)\Big|_1^2 \qquad [I = 3t^2 + 4t^3]$$
$\therefore q = 22$C

9. (2) Given $R_{100} = 10\Omega$, $R_o = 8\Omega$
$\therefore R_{100} = R_0(1+\alpha\Delta T)$
$$\Rightarrow 10 = 8(1+\alpha\times100) \Rightarrow 100\alpha = \frac{1}{4}$$
Also, $R_{100} = R_0(1+\alpha\Delta T)$
$\Rightarrow R_{400} = 8(1+400\alpha) = 8(1+1) = 16\Omega$

10. (32)
$$\because R = \frac{\rho\ell}{A} = \frac{\rho V}{A^2} \Rightarrow R \propto \frac{1}{r^4}$$
$$\therefore \frac{R_A}{R_B} = \frac{r_B^4}{r_A^4}$$
$$\Rightarrow \frac{R_A}{2} = \left[\frac{4\times10^{-3}}{2\times10^{-3}}\right]^4 \Rightarrow R_A = 32\Omega$$

11. (16)
From $R = \frac{\rho l}{A}$, $R \propto \frac{l}{r^2}$
On starching the wire, the length increases, but radius decreases to keep the volume constant, therefore
initial volume = final volume
$$\pi r^2 l = \pi\frac{r^2}{4}l'$$
$l' = 4l$
$$\frac{R_1}{R_2} = \left(\frac{4l}{\frac{r^2}{4}}\right)\frac{r^2}{l} = 16$$
$R_1 = 16\,R \Rightarrow x = 16$

12. [748]
Resistance of the wire,
$R = R_0(1 + \alpha\Delta T)$
$$\frac{\Delta R}{R_0} = \alpha\Delta T$$
Case-I 0°C → 100°C
$$\frac{10.2-10}{10} = \alpha(100-0) \qquad ...(i)$$
Case-II 0°C → T°C
$$\frac{10.95-10}{10} = \alpha(t-0) \qquad ...(ii)$$
$T = 475 + 273 = 748$ K

13. (1027) We have given the resistance of the element in 50Ω and
$\alpha = 2.4 \times 10^{-4}$°C^{-1}
R = R0(1 + αΔT)
$62 = 50\,[1 + 2.4 \times 10^{-4}\Delta T]$
ΔT = 1000°C
⇒ T – 27° = 1000°C
T =1027°C

14. [300] Volume of wire will remain constant
$$\Rightarrow \pi r_1^2\ell_1 = \pi r_2^2(2\ell_1)$$
$$r_2 = \frac{r_1}{\sqrt{2}}$$
$$R_2 = \frac{\rho\ell_2}{\pi r_2^2} = \frac{\rho 2\ell_1}{\pi\left(\frac{r_1}{\sqrt{2}}\right)^2}$$
$$R_2 = 4\frac{rl_1}{pr_1^2}$$
$R_2 = 4R_1$
$$\% \text{ Change in Resistance} = \frac{\Delta R}{R}\times100$$
$$= \frac{4R_1 - R_1}{R_1}\cdot 100 = 300\%$$

15. (*c*) Volume = constant
$A\ell = C$
$$\therefore \frac{d\ell}{\ell} + \frac{dA}{A} = 0$$
$$\Rightarrow \frac{dA}{A} = \frac{-d\ell}{\ell} = -0.4\%$$
Now, $R = \frac{\rho\ell}{A}$
$$\therefore \frac{dR}{R}\times100 = \left(\frac{d\ell}{\ell} - \frac{dA}{A}\right)\times100$$
$= 0.4 - (-0.4) = 0.8\%$

16. [48]
$I = A \times J$
$$= \pi\left(R^2 - \frac{R^2}{4}\right)J$$
$$= \frac{3\pi R^2}{4}\times J$$
$$= \frac{3\pi\times\left(4\times10^{-3}\right)^2}{4}\times4\times10^6$$
$= 48\pi$

17. [12]
Given, Current density $J = 1 \times 10^6$ A/m²
$r = 4$ mm $= 4 \times 10^{-3}$ m
$$r_1 = \frac{r}{2}$$
Formula of current density,
$$J = \frac{I}{A} = \frac{I}{\pi r^2 - \pi(r')^2}$$
$$\therefore I = \pi\left(r^2 - \frac{r^2}{4}\right)\times J$$
$$= \pi\times\frac{3r^2}{4}\times1\times10^6$$
$$= \pi\times\frac{3}{4}\times16\times10^{-6}\times10^6$$
$= 12\,\pi$
$\Rightarrow x = 12$

18. (*a*) Using formula, $R_T = R_o[1 + \alpha(T - T_o)]$
If T = 10°C and $R_T = 2\ \Omega$ then,
$2 = R_o[1 + \alpha(10 - T_o)]$ ...(*i*)
If T = 30°C and $R_T = 3\ \Omega$
$3 = R_o[1 + \alpha(30 - T_o)]$ ...(*ii*)
Solving (*i*) and (*ii*), we get
$$\alpha = \frac{1}{30} = 0.033$$

19. [144]
We know that, $R = \frac{V}{I} = 1\Omega$
$$R = \frac{\rho l}{A} \Rightarrow 1 = \frac{x\times10^{-3}\times31.4}{\pi\times(1.2)^2}$$
$\Rightarrow x = 144$

20. (*a*) $R_1 = \rho\frac{L_1}{A_1}$
Vol. $V = Al$
$A_1l_1 = A_2(3L_1)$
$$\Rightarrow A_2 = \frac{A_1}{3}$$
So, $R_2 = \rho\left(\frac{3L_1}{A_1/3}\right) = 9\rho\frac{L_1}{A_1}$
$$\therefore \frac{R_2}{R_1} = 9$$

21. (*a*) Fact based.

22. (*b*) Let *x* be the length of *x*.
∴ Length of *Y* = 1 – *x*
And, Rw = 2Ry
$$\Rightarrow \rho\left(\frac{2x}{A/2}\right) = \frac{2\rho(1-x)}{A}$$
$$\Rightarrow x = \frac{1}{3}$$
$$\therefore \frac{L_x}{L_y} = \frac{1/3}{1-1/3} = \frac{1}{2}$$

23. (*d*) $i = \frac{dQ}{dt}$
$dQ = idt$
$$\int_0^Q dQ = \int_0^t i\,dt$$
$$Q = \int_0^{15}\left(\alpha_0 t + \beta t^2\right)dt$$
$$= \left[\alpha_0\frac{t^2}{2} + \beta\frac{t^3}{3}\right]_{t=15} - \left[\alpha_0\frac{t^2}{2} + \beta\frac{t^3}{3}\right]_{t=0}$$
$$= \left[20\times\frac{15\times15}{2} + 8\times\frac{15\times15\times15}{3}\right] - 0$$
$= 2250 + 9000$
$= 11250$ C

24. [5] Given, radius of cylindrical wire, $r = 0.5$ mm, conductivity, $\sigma = 5 \times 10^7$ s/m, Electric field strength, $\vec{E} = 10\times10^{-3}$ V/m $= 10^{-2}$ V/m
As we know,
$\therefore I = \sigma EA = 5 \times 10^7 \times 10^{-2} \times \pi \times (0.5 \times 10^{-3})^2$
$= 125\,\pi$mA $= x^3\pi$mA
$\therefore x = 5$

25. (*d*) Relation between current and drift velocity.
$i = neAV_d$
Number of free ens in per-cubic
$$n = \frac{i}{eAV_d}$$
$$= \frac{10}{1.6\times10^{-19}\times5\times10^{-6}\times2\times10^{-3}}$$
$= 625\times10^{25}$

26. (c) $\vec{J}\cdot\vec{A} = i$

$JA\cos\theta = i$

$J = \frac{E}{\rho} = \frac{i}{A\cos\theta}$

$E = \frac{i\rho}{A\cos\theta} = \frac{5\times44\times10^{-8}}{0.04\times\left(\frac{1}{2}\right)}$

$= \frac{5\times44\times10^{-8}\times2}{4\times10^{-2}}$

$E = 110\times10^{-6} = 11\times10^{-5}\left(\frac{V}{m}\right)$

27. (c) Current throughout the conductor would be constant.

Since, $Vd \propto E$ & $E \propto \frac{1}{r^2}$

So, E increases with decrease in the radius.

Also $Vd \propto E$ So, drift speed increases.

28. (a) Resistance of the conductor

$R = R_0[1 + \alpha\Delta T]$

$\Rightarrow 16 = R_0[1 + \alpha(15 - 0)]$

$20 = R_0[1 + \alpha(100 - 0)]$

$\Rightarrow \frac{16}{20} = \frac{1+\alpha\times15}{1+\alpha\times100}$

$\Rightarrow \alpha = 0.003°C^{-1}$

29. (b) Here, resistivity of the four material are given as follows:

$\rho_{ni} = 9.8 \times 10\text{-}8$

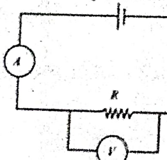
$\rho_A = 2.65 \times 10\text{-}8$

$\rho_c = 1.724 \times 10\text{-}8$

$\rho_g = 5.65 \times 10\text{-}8$

Hence, $\rho_{ni} > \rho_A > \rho_c$

30. (c) In Ohm's law experiment, ammeter is connected in series because in series same current will flow through it. But voltmeter is connected in parallel to resistor to measure the potential difference across it.



31. (a) Current flows in copper wire,

$I = ne\,AV_d$

$32.5 = 9 \times 1023 \times 1.6 \times 10\text{-}19 \times 5 \times Vd$

$V_d = 0.02$

32. (d) Resistivity, $\rho = \frac{2m}{ne^2\tau}$

$\rho = \frac{9.1\times10^{-31}}{(25\times10^{-15})\times(1.6\times10^{-19})\times(8.5\times10^{28})}$

$= 1.67\times10^{-8}\,\Omega m$

33. (d) Here, we have given resistance as a function of temperature, $R(T)$

$\frac{\frac{1}{T^2}}{\frac{1}{T_0^2}} + \frac{\ln R(T)}{\ln R(T_0)} = 1$

Now,

$\Rightarrow \ln R(T) = \ln R(T_0)\left(1 - \frac{T_0^2}{T^2}\right)$

$R(T) = R(T_0)e^{\left(\frac{T_0}{T}\right)^2}$

Hence, option (d) is correct answer.

34. (b) Mobility, $\mu = \frac{V_d}{E}$ and $E = \rho J$

$m = \frac{1.1\times10^{-3}}{1.7\times10^{-8}\times\frac{5}{\pi\times25\times10^{-6}}} \approx 1.01\ m^2/Vs$

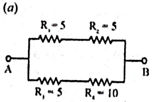
$H = \frac{1.1\times10^{-3}\times\pi\times25\times10^{-6}}{1.7\times10^{-8}\times5} \approx 1.01\ m^2/Vs$

35. (d) Resistance between the two sphere

$R = \int_a^b \frac{\rho dx}{4\pi x^2}$

$= \frac{\rho}{4\pi}\left(\frac{1}{a} - \frac{1}{b}\right)$

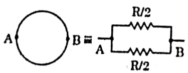
36. (a)



Option (a) is the best fit, because

$\frac{1}{R_P} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{1}{6}$

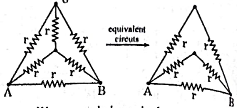
37. (None)



$R_{wire} = \frac{\rho L}{A} = \frac{2\times10^{-6}\times25}{5\times10^{-6}} = 10$

$R_{eq} = \frac{R}{4} = \frac{10}{4} = 2.5\Omega$

38. (d) Resistance of each resistor will be $r = \frac{R}{6}$



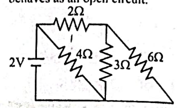
We can see balanced wheat stone bridge in uppar circuit.

equivalent resistance between A and B

$\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{2r} + \frac{1}{r} = \frac{2}{r}$

$r_{eq} = \frac{R}{12}$

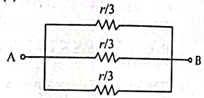
39. [1] Assuming DC Analysis Capacitor behaves as an open circuit.



$R_{eq} = 2\Omega$

$\frac{1}{2}I = \frac{2}{2} = 1A$

40. (b)

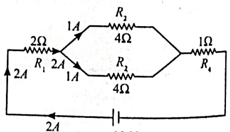


$R_{AB} = \frac{r/3}{3} = \frac{r}{9}$

41. (a) Each part have resistance $= \frac{R}{5}$

Thus Req. $= \frac{1}{5}\times\frac{R}{5} = \frac{R}{25}$

42. (a)



$R_{eq} = 2\Omega + 2\Omega + 1\Omega = 5\Omega$

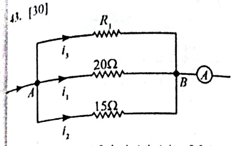
$i = \frac{V}{R_{eq}} = \frac{10}{5} = 2A$

Current in resistance $R_3 = 2 \times \left(\frac{4}{4+4}\right)$

$= 2\times\frac{4}{8} = 1A$

Hence, option (b) is correct answer.

43. [30]



Given, $i_1 = 0.3$ A, $i_1 + i_2 + i_3 = 0.9$ A

So, $V_{AB} = i_1 \times 20\,\Omega = 20 \times 0.3$ V $= 6$ V

$i_2 = \frac{6V}{15W} = \frac{2}{5}A$

$i_1 + i_2 + i_3 = \frac{9}{10}A$

$\frac{3}{10} + \frac{2}{5} + i_3 = \frac{9}{10}$

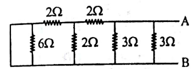
$i_3 = 0.2$ A

So, $i_3 \times R_1 = 6V$

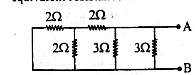
$(0.2)\,R_1 = 6$

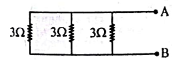
$R_1 = \frac{6}{0.2} = 30\Omega$

44. [1]



Resistance 6 ohm will get short circuit, equivalent resistance is





$R_{eq} = 3\times\frac{1}{3} = 1\Omega$

45. [5] 10 equal part imply

Each part has resistance = 2Ω

2 parts are connected in parallel

so, for parallel contribution

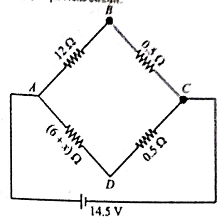
$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{2}$

$R_{eq} = 1\Omega$

Now, there will be 5 parts each of resistance 1 Ω, they are connected in series.

$R_{eq} = 5R,\ R_{eq} = 5\Omega$

46. (c) Equivlent circuit:



Wheatstone Bridge is balanced, therefore,

$\frac{12}{6+x} = \frac{0.5}{0.5}$

$x = 6\,\Omega$

47. (d) Here R_2, R_3, R_4 are in parallel

For parallel combination of resistors,

$\frac{1}{R_{eq}} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$

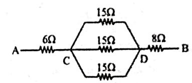
$R_{234} = 2\Omega$

R_{234} is in series with R_1 so

$R_{eq} = R_{234} + R_1 = 2 + 10 = 12\Omega$

$i = \frac{12}{12} = 1\ Amp$

48. (d) Equivalent circuit will become



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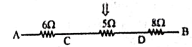
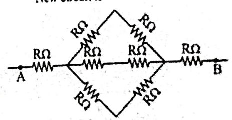


$\Rightarrow R_{eq} = 6\Omega + 5\Omega + 8\Omega = 19\Omega$

49. (b) From symmetry we can remove two middle resistance.

New circuit is



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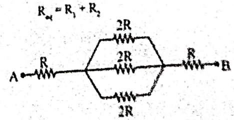
For series combination of resistor

$R_{eq} = R_1 + R_2$



For parallel combination of resistor

$R_{eq} = \frac{1}{R_1} + \frac{1}{R_2}$

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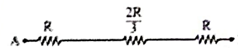

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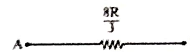
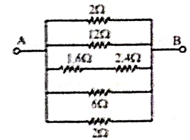


50. (a) Equivalent circuit can be drawn as



$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{12} + \frac{1}{4} + \frac{1}{6} + \frac{1}{2} = \frac{3}{2}$

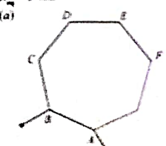
$\Rightarrow R_{eq} = \frac{2}{3}\Omega$

51. (a) All the resistors are connected in parallel, therefore

$\frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{20} + \frac{1}{20}$

$R_{eq} = 5\ k\Omega$

52. (a)



Let resistance of each arm is r.

$\Rightarrow r = R/n$

$R_{eq(AB)} = \frac{R_1R_2}{R_1 + R_2}$

$R_{eq} = \frac{r(n-1)r}{r+(n-1)r}$ $\quad R_{eq} = \frac{r(n-1)r}{nr}$

$\Rightarrow R_{eq} = \frac{(n-1)R}{n^2}$ $\quad \left[\because r = \frac{R}{n}\right]$

53. (b) $R_{eq} = R_1 + R_2 + R_3$

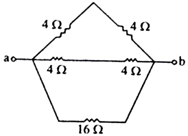
In series, Req is greater than the any resistor present in the circuit

So, Statement-I is false.

Resistivity depends on temperature.

So, Statement-II is False

54. (b) Using balanced wheat stones bridge (in 4 Ω network)



⇒ $R_{eq} = \frac{16\times4}{16+4} = \frac{16}{5}\Omega = 3.2\,\Omega$

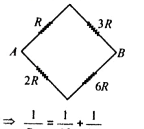
55. (d) Equivalent resistance of circuit $R_{eq} = 3+1+2+4+2 = 12\,\Omega$

Current through the battery is $i = \frac{24}{12} = 2A$

$I_4 = \frac{R_5}{R_4+R_5}\times 2 = \frac{5}{20+5}\times 2 = \frac{2}{5}A$

and, $I_5 = 2 - \frac{2}{5} = \frac{8}{5}A$

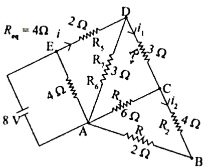
56. (d) As the wheatstone bridge is in balanced condition, no current will flow through the 9R.



⇒ $\frac{1}{R_{eq}} = \frac{1}{4R}+\frac{1}{8R}$

⇒ $R_{eq} = \frac{8R}{3}$

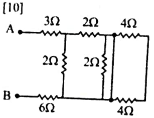
57.(d)



$i = \frac{8}{4} = 2A$; $i_1 = \frac{2\times3}{3+6} = \frac{2}{3}A$

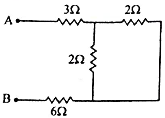
$i_2 = \frac{2/3}{2} = \frac{1}{3}A$

58. [10]



Both 4Ω resistance gets short due to the conducting wire parallel to 2Ω.

Removing the 4Ω resistors that have no current.



$R_{eq} = 3 + (2 \| 2) + 6 = 3 + 1 + 6$

⇒$R_{eq} = 10\Omega$

59. [2] $\frac{2}{3} = \frac{\frac{x}{x+1}}{x}$

⇒ $\frac{2}{3} = \frac{1}{x+1}$ ⇒ $x = 0.5 = \frac{1}{2}$

$n = 2$

60. [100] Series combination gives maximum resistance whereas parallel combination gives minimum resistance.

$R_{max} = 10R$, $R_{min} = \frac{R}{10}$

∴ $\frac{R_{max}}{R_{min}} = \frac{10R}{R/10} = 100$

61. (b) $R = R_1 + R_2$

⇒ $\frac{2}{\sigma A} = \frac{1}{\sigma_1 A}+\frac{1}{\sigma_2 A} \Rightarrow \sigma = \frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

62. [4] Formula of resistance, $R = \frac{rl}{A}$

Equivalent resistance of eight wire in parallel is R/8

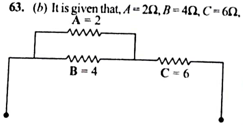
According to question,

$R = \frac{\rho\ell}{\pi\left(\frac{d}{2}\right)^2} \Rightarrow R \propto \frac{\ell}{d^2}$

$\frac{R_1}{R_2} = \frac{l_1}{l_2}\cdot\frac{d_2^2}{d_1^2}$

$\frac{R}{R} = \frac{1}{2}\cdot\frac{d^2}{d^2} \Rightarrow d_2 = 4d$

63. (b) It is given that, $A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$.



From the diagram given above,

$R_{eq} = \frac{2\times4}{2+4}+6 = \frac{8}{6}+6 = \frac{4}{3}+6$

∴ $R_{eq} = \frac{22}{3}\Omega$

64. (a) Arrangement of the resistens is equivalent to wheat stone bridge. Therefore equivalent resistance of the circuit.

$R_{eq} = \frac{4\times4}{4+4}+\frac{4\times4}{4+4} = 4\Omega$

From ohm's law

$I = \frac{V}{R} = \frac{40}{4} = 10A$

65. (a) Statement-I

$80 = \frac{\rho l}{A}$

$R' = \frac{\rho l}{4A} = \frac{80}{4} = 20\Omega$

So, $R_{eq} = \frac{R'}{4} = \frac{20}{4} = 5\Omega$

Statement-II

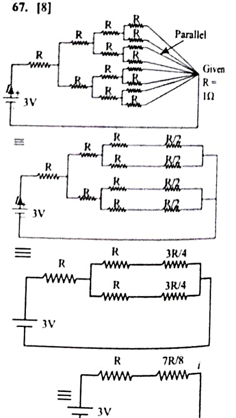
$\frac{P_1}{P_2} = \frac{V^2}{2R}\times\frac{3R}{V^2} = \frac{3}{2}$

66. (a) The circuit is balanced whetstone bridge. So there is no current in 5Ω resistor.

⇒ $R_{eq} = \frac{6\times12}{6+12}+2 = 6\Omega$

⇒ $I = \frac{V}{R_{eq}} = \frac{6}{6} = 1A$

67. [8]



Current, $I = \frac{3\times8}{15R} = \frac{8}{5R}$

On comparing above result with $I = a/5$, we get

$a = 8$

68. [2] Equivalent resistance of parallel combination of three resistors = 3 ohm, I = 6/3 = 2A

69. [3] Equivalent resistance of the circuit

$Req = \frac{ma}{3}+\frac{a}{2m}$

$\frac{dR_{eq}}{dm} = 0 \Rightarrow \frac{a}{3}-\frac{a}{2}m^2 = 0$

$m^2 = \frac{3}{2} \Rightarrow m = \sqrt{\frac{3}{2}}$

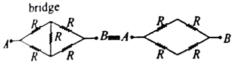
$x = 3$

70. (c) Using hit and trial method, from option (c) we get,

$R_{eq} = \frac{2\times4}{2+4}+6+8 = \frac{46}{3}\Omega$

71. (b) As, 5 Ω is connected across conductor so we can remove it. $Req = \frac{3\times\frac{3}{2}}{3+\frac{3}{2}} = 1\Omega$

72. (b) Given circuit can be re-drawn and it becomes case of balanced Wheatstone bridge



$R_{AB} = \frac{(2R)(2R)}{2R+2R} = R$

73. (a) $R_1 = \rho_1\frac{l}{A}$ $R_2 = \rho_2\frac{l}{A}$

Equivalent resistance in parallel form.

$\frac{1}{R} = \frac{1}{R_1}+\frac{1}{R_2} = \frac{A}{l}\left[\frac{1}{\rho_1}+\frac{1}{\rho_2}\right]$

$= \frac{3\times10^{-6}}{1}\left[\frac{1}{1.7\times10^{-8}}+\frac{1}{2.6\times10^{-8}}\right]$

$= 12\times10^{-2}\left(\frac{10}{17}+\frac{10}{26}\right)$

$R = 0.858\times10^{-3}\,\Omega$

74. (d) Formula of resistance, $R = \frac{\rho l}{A}$

$\rho_1 = 12\mu$ cm

$\rho_2 = 51\mu$ cm

Equivalent resistance, $\frac{R_1R_2}{R_1+R_2} = 3$

$\frac{\left(\rho_1\frac{l}{A}\right)\left(\rho_2\frac{l}{A}\right)}{\left(\frac{\rho l}{A}\right)+\left(\frac{\rho_2 l}{A}\right)} = 3$

$l = \frac{3A(\rho_1+\rho_1)}{\rho_1\rho_2}$

$= \frac{3\times\left[\frac{\pi}{4}\times\left(2\times10^{-3}\right)^2\right][12+51]\times10^{-6}\times10^{-2}}{12\times10^{-6}\times10^{-2}\times51\times10^{-6}\times10^{-2}}$

= 97m

75. (b) $\frac{1}{R_{eq}} = \frac{1}{R_1}+\frac{1}{R_2} = \frac{1}{4}+\frac{1}{4}$

⇒ Req = 2

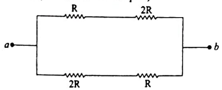
$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2}+\frac{\Delta R_2}{R_2^2}$

$\frac{\Delta R_{eq}}{4} = \frac{0.8}{16}+\frac{0.4}{16} = \frac{1.2}{16}$

So, $\Delta Req = 0.3$

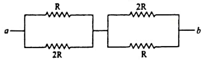
∴ $Req = (2 \pm 0.3)\Omega$

76. [9] When switch is open,



$R_{ab} = \frac{3R}{2}$

When switch is closed,



$R'_{ab} = 2\times\frac{R\times2R}{R+2R} = \frac{4R}{3}$

$\frac{R_{ab}}{R'_{ab}} = \frac{3R/2}{4R/3} = \frac{9}{8} = \frac{x}{8} \Rightarrow x = 9$

77. [6] Equivalent resistance, $Req = 6\ \Omega$

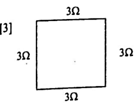
Current through the battery,

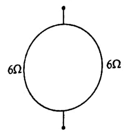
$i = \frac{12V}{6\Omega} = 2A$

Voltage across, 15 Ω resistance, V (15Ω)

= 6 V

78. [3]





$R_{AB} = \frac{6\times6}{12} = 3$

79. [3] $Req = 5+1+1 = 7k\Omega$

∴ $I = \frac{V_{eq}}{R_{eq}} = \frac{21}{7\times10^3} = 3mA$

80. [70]

$R_{eq} = 10+\frac{50\times20}{50+20} = \frac{170}{7}\Omega$

⇒ $i = \frac{170}{\frac{170}{7}} = 7A$

⇒ $x = 10\times7 = 70$ V

81. [2] Using KCL at P, $i_1 + i_2 = 6$ ...(i)

and using KVL in closed circuit $PQRP$,

Solving above equations, we get, $i_1 = 2A$.

82. [20]

In series connection, $R_{eq} = nR = 10n$

⇒ $I_{se} = \frac{20}{10+10n} = \frac{2}{1+n}$

In parallel connection, $R_{eq} = \frac{10}{n}$

⇒ $I_p = \frac{20}{\frac{10}{n}+10} = \frac{2n}{1+n}$

According to the question,

⇒ $\frac{I_p}{I_{se}} = 20 = \frac{2n/1+n}{2/1+n} \Rightarrow n = 20$

83. [4] In parallel combination,

$R_{eq} = \frac{R_1R_2}{R_1+R_2}$

⇒ $\frac{\rho\ell}{2A} = \frac{\rho_1\frac{\ell}{A}\times\rho_2\frac{\ell}{A}}{\rho_1\frac{\ell}{A}+\rho_2\frac{\ell}{A}}$ $\left(\because R = \rho\frac{l}{A}\right)$

⇒ $\frac{\rho}{2} = \frac{6\times3}{6+3} = 2$

⇒ $\rho = 2\times2 = 4\Omega-$cm

84. [4] $s = np$

$R_1+R_2 = n\left[\frac{R_1R_2}{R_1+R_2}\right]$

⇒ $R_1^2+R_2^2+2R_1R_2 = nR_1R_2$

⇒ $R_1^2+(2-n)R_1R_2+R_2^2 = 0$

For real roots, $b^2-4ac\ge0$

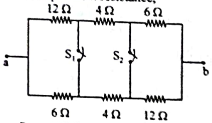
$[(2-n)R_2]^2-4\times1\times R_2^2\ge0$

⇒ $(2-4)^2R_2^2\ge4R_2^2 \Rightarrow 2\times n\ge\pm2$

⇒ $2-n\ge-2 \Rightarrow n\ge4$

So, minimum value for $n = 4$

85. [10]
The equivalent resistance,



$R_{eq} = (12\|6) + (4\|4) + (4\|12)$

$= \left(\frac{12\times6}{18}\right)+\left(\frac{4\times4}{8}\right)+\left(\frac{6\times12}{18}\right)$

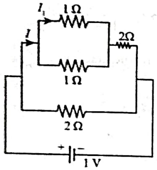
$=4+2+4=10\Omega$

86. (c) Resistance in parallel

$=\frac{R_1R_2}{R_1+R_2}=\frac{1}{2}\Omega$

Now, resistance in series R'

$=\frac{1}{2}+2=2.5\Omega$

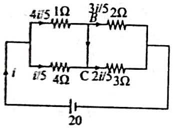


Current through upper breach

$I=\frac{1}{2.5}=0.4A$

Current through 1Ω $(I_1)=\frac{I}{2}=0.2A$

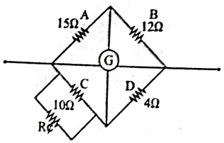
87. (b) $R_{eff}=\frac{4}{5}+\frac{6}{5}=2\Omega$



$i=\frac{20}{2}=10A$

$I=\frac{4i}{5}-\frac{3i}{5}=\frac{i}{5}=2A$

88. [10]



Wheatstone bridge balance conditions

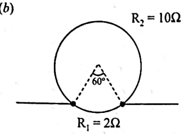
$\frac{A}{B}=\frac{C}{D}$

$\Rightarrow \frac{10R'}{10+R'}\times12=15\times4$

On solving

R' = 10 Ω

89. (b)



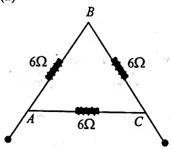
$\Rightarrow R=\frac{\rho\ell}{A}=\frac{\rho\ell^2}{V}\quad R\propto \ell^2$

$R=12\Omega$ (new resistance of wire)

$R_1=2\Omega, R_2=10\Omega$

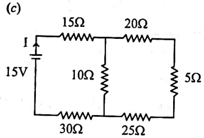
$R_{eq}=\frac{10\times2}{10+2}=\frac{5}{3}\Omega$

90. (a)



So, $R_{AC}=\frac{12\times6}{18}=4\Omega$

91. (c)

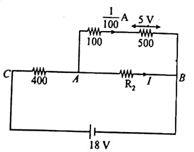


Req = 15 + {(20 + 5 + 25) || 10} + 30

$Req=15+\frac{25}{3}+30=\frac{160}{3}$

Current, $I=\frac{E}{R_{eq}}=\frac{15}{\left(\frac{160}{3}\right)}=\frac{9}{32}$ amp

92. (a)



$V_{AB}=\left(\frac{1}{100}\right)600=6$ volt

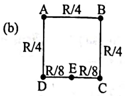
$\Rightarrow V_{AC}=18V-6V=12$ volt

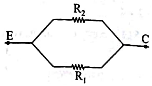
$I_{AC}=\frac{12}{400}=\frac{3}{100}\Rightarrow I=I_{AC}-\frac{1}{100}$

$\Rightarrow I=\frac{3}{100}-\frac{1}{100}=\frac{2}{100}$

$\Rightarrow R_2=\frac{V_{AB}}{I}=\frac{6\times100}{2}=300\Omega$

93. (b)





$R_1=\frac{R}{8}, R_2=\frac{R}{8}+\frac{R}{4}+\frac{R}{4}+\frac{R}{4}+\frac{R}{4}=\frac{7R}{8}$

$\frac{1}{R_{EC}}=\frac{1}{R_1}+\frac{1}{R_2}$

$\frac{1}{R_{EC}}=\frac{8}{7R}+\frac{8}{R}$

$REC=\frac{7R}{8+56}=\frac{7R}{64}$

94. (c) Resistance of each bulb is:

$R=\frac{V_{total}^2}{P}$ (from $P=\frac{V_{total}^2}{R}$ when alone),

Total resistance, when resistors are connected in series:

$Rs=nR=n\cdot\frac{V_{total}^2}{P}$

Total power: $P_s=\frac{V_{total}^2}{R_s}=\frac{V_{total}^2}{n\cdot\frac{V_{total}^2}{P}}=\frac{P}{n}$

95. (c) Given: Voltage V = 100V, Current I = 1 A

So, total input power.

$P_{in}=V\times I=100W$

Efficiency is: $\eta=\frac{P_{useful}}{P_{in}}=0.916$

Thus, useful output power: $P_{useful}=0.916\times100=91.6W$

Power lost: Loss $=P_{in}-P_{useful}=100-91.6=8.4W=2$ cal/s

96. (d) $\frac{v^2}{R}=W$ ...(i)

When the wire is cut into two halves

$\frac{v}{\frac{1}{2}\left(\frac{R}{2}\right)}=W'$ ...(ii)

From (i) & (ii), we get

W' = 4W

97. (c)Power, $P = i^2R$

$P_{int}=I_{int}^2R$

$P_{final}=(0.8I_{int})^2R$

% change in power =

$=\frac{P_{final}-P_{int}}{P_{int}}\times100$

$=(0.64-1)\times100=-36\%$

98. [3]Equivalent Resistance,

$R_{eq}=\frac{4}{3}\Omega$

We know, $P=\frac{V^2}{R_{eq}}=\frac{4}{4/3}=3\Omega$

99. (a) Resistance,

$R=\frac{V^2}{P}=\frac{(200)^2}{50}=\frac{40000}{50}$

R = 800 Ω

Power dissipated

$P=\frac{(V_{applied})^2}{R}=\frac{(100)^2}{800}=12.5$ watt

100. (b) The current through 5Ω resistor is i_1 and 10 Ω resistor is i_2

$\frac{i_1}{i_2}=\frac{10}{5}=\frac{2}{1}$

$\frac{P_1}{P_2}=\frac{i_1^2R_1}{i_2^2R_2}=\left(\frac{2}{1}\right)^2\times\frac{5}{10}=\frac{2}{1}$

101. (c) Resistance of the material, $R=\frac{\rho\ell}{A}$,

Now, $P=\frac{v^2}{R}\Rightarrow P\propto\frac{1}{\ell}$

$\Rightarrow\frac{P_1}{P_2}=\frac{t_2}{t_1}=\frac{15}{20}=\frac{\ell_2}{\ell_1}$

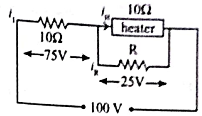
$\therefore \ell_2=\frac{3}{4}\ell_1$

102. [5] Resistance of the heater,

$R_{heater}=\frac{V^2}{P}=\frac{(100)^2}{1000}=10\Omega$

For heater, $P_{heater}=\frac{V^2}{R}\Rightarrow V=\sqrt{PR}$

$V=\sqrt{62.5\times10}\Rightarrow V=25v$



$i_1=\frac{75}{10}=7.5A, i_H=\frac{25}{10}=2.5A$

$i_R=i_1-i_H=5$

Using ohm' law, V = IR

$\therefore R=\frac{25}{5}=5\Omega$

103. (b) $H=I^2Rt=4^2R\times10=160R$

$H'=(I')^2Rt=16^2R\times10=2560R=16H$

104. (d) The formula of power $P=I^2R_{eq}$

$\Rightarrow P\propto Req$

$R_A=\frac{3R}{2},R_B=\frac{2R}{3},R_C=\frac{R}{3},R_D=3R$

105. (a) Heat energy produced in a resistor R connected to potential difference V is

$H=\frac{V^2}{R}\times t$

As the resistors are connected in parallel

$\Rightarrow\frac{H_1}{H_2}=\frac{\frac{V^2t}{R}}{\frac{V^2t}{3R}}=3:1$

106. (a) Parallel combination, Req = $\frac{R}{2}$

Heat produced

$H_p=\left[\frac{V^2}{\left(\frac{R}{2}\right)}\right]t=\frac{2V^2t}{R}$ ...(i)

Series combination, $R_{eq}=2R$

Heat produced $H_s=\left(\frac{V^2}{2R}\right)t$ ...(ii)

$\therefore\frac{H_p}{H_s}=4$ [Dividing (i) and (ii)]

107. [16]

Initial $P_i=\frac{V_0^2}{R}$

After cutting wire into two half

$R'=\frac{R}{2}$

$\because R=\frac{\rho l}{A}$

$\therefore P_f=\frac{V_0^2}{R'}\times2=\frac{V_0^2}{R}\times4\Rightarrow P_f=P_i\times4$

$\frac{P_f}{P_i}=4=\frac{\sqrt{x}}{1}\Rightarrow x=16$

108. [14]

$\frac{v_1}{v_2}=\frac{R_1}{R_2}=\frac{P_2}{P_1}=\frac{60}{100}=\frac{3}{5}$

$v_1=\frac{3}{8}\times220=3\times27.5=82.5V$

$\therefore P_1'=\left(\frac{V_1}{V}\right)^2P=\left(\frac{330}{4\times220}\right)^2\times100=14W$

109. [450]

$H=i^2Rt$

$300=2^2\times R\times15$

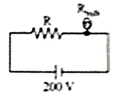
$\Rightarrow R=300/60=5\Omega$

Now, for $i=3A, t=10s, R=5\Omega$

$H=3^2\times5\times10=450J$

110. (a) $R_{bulb}=\frac{V^2}{P_{bulb}}=\frac{100\times100}{500}=20\Omega$

$I=\frac{V_{supply}}{R_{bulb}+R}=\frac{200}{20+R}$



According to the question.

Power delivered by bulb = 500 W

$\Rightarrow I^2 R_{bulb} = 500$

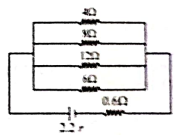
$\Rightarrow\left(\frac{200}{20+R}\right)^2\times20=500$

$\Rightarrow(R+20)^2=\frac{200\times200\times20}{500}=1600$

⇒ R + 20 = 40

∴ R = 20 Ω

111. (a) Equivalent resistance of the circuit.



$=0.6\Omega+\frac{3\times\frac{48}{14}}{3+\frac{48}{14}}\Omega$

$=0.6\Omega+\frac{(144/14)\Omega}{\left(\frac{90}{14}\right)}$

$=0.6\Omega+1.6\Omega=2.2\Omega$

Power dissipated in the circuit

$P=\frac{V^2}{R_{eq}}=\frac{(2.2V)^2}{2.2\Omega}=2.2W$

$\Delta U = \Delta Q - \Delta W$

$\frac{\Delta U}{\Delta t} = \frac{\Delta Q}{\Delta t} - \frac{\Delta W}{\Delta t} = \frac{6000}{60} - 90 = 10\text{J/s}$

$t = \frac{\Delta U}{10} = \frac{2.5\times10^3}{10} = 2.5\times10^2$ seconds

113. (a) Using Joule's law of heating, heat generated in the resistance

$H = i^2RT$

$H_1 = 500 = (1.5)^2 R(20)$

$H_2 = (3)^2 R(20) \Rightarrow H_2 = H_1\times\left(\frac{3}{1.5}\right)^2$

$= 500\times\frac{9}{2.25} = 2000$ J

114. [3840] $E = I^2Rt$

$\Rightarrow 192 = 4^2R(1)$

$\Rightarrow R = 12\,\Omega$

As current is doubled and time = 5s

So, energy dissipated in this case

$E' = 3840$ J

115. [4] In first condition $R_1 = 36\Omega$

In second condition $R_2 = 18\Omega$

$P_1 = \frac{V^2}{R_1} = \frac{(240)^2}{36}$

$P_2 = \frac{V^2}{R_2} + \frac{V^2}{R_2} = \frac{(240)^2}{18} + \frac{(240)^2}{18}$

$P_2 = \frac{(240)^2}{9}$

$\frac{P_1}{P_2} = \frac{(240)^2/36}{(240)^2/9} = \frac{1}{4} \quad x = 4$

116. [2500] Energy dissipated by a resistor

$H = i^2 Rt$

$R = \frac{10\times10^{-3}}{(2\times10^{-3})^2\times1} = 2500\Omega$

Hence, the resistance is 2500 W

117. [50]

Given, $P = 20W$, Supply voltage, $V_s = 200V$

Resistance of electric Bulb,

$R_B = \frac{V^2}{P} = \frac{(100)^2}{200} = 50\Omega$

Formula of power,

$P = I^2(R_B)$

$= \left(\frac{V_s}{R+R_B}\right)^2 .R_B$

$\Rightarrow \left(\frac{200}{R+50}\right)^2\times50 = 200$

$R = 50\Omega$

118. (b) Total power consumed (P) is $= (15\times45) + (15\times100) + (15\times10) + (2\times1000) = 4325$ W.

So, current (I)

$= \frac{\text{Power consumed}}{\text{Voltage supplied}} = \frac{4325}{220} = 19.66A$

Hence, answer is 20 Amp.

119. (c) Power in external $R = \left(\frac{\varepsilon}{r+R}\right)^2 R$

This power is maximum, when $R = r$

$r = \int_a^b \rho\frac{dx}{2\pi xl}$

$r = \frac{\rho}{2\pi l}\ln\frac{b}{a}$

120. (a) $I^2R = 0.5$

$iR = 2.5$

$\Rightarrow i = 0.2$ A and

$R = 12.5\,\Omega$

$V = \varepsilon - ir$

$2.5 = 3.0 - 0.2r$

$\Rightarrow r = 2.5\,\Omega$

Power dissipated in internal resistance

$P_r = \frac{P_R}{R/r} = \frac{0.5}{5} = 0.1\text{W} = i^2r = 0.1\ W$

121. (a) $R_{eq} = (4R\|4R) + R + (6R\|12R) + R$

$R_{eq} = 8R$

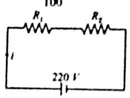
$\because p = \frac{V^2}{R_{eq}} = \frac{(16)^2}{8R} = 4$

$\therefore R = 8\Omega$

122. (a) Resistance, $R = \frac{V^2}{P}$

$\Rightarrow R_1 = \frac{(200)^2}{25} = 1936\,\Omega$

$R_2 = \frac{(200)^2}{100} = 484\,\Omega$



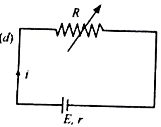
$i = \frac{220}{R_1+R_2} = \frac{1}{11}A = 0.09A$

Dissipated powers,

$P_1 = i^2R_1 = (0.09)^2\,1936 = 16W$

$P_2 = i^2R_2 = (0.09)^2\,484 = 4W$

123. (d)



Internal resistance = r

External resistance = R

For maximum power delivered by cell

Internal resistance = External resistance

$\Rightarrow r = R$

124. (a) $R = \frac{P}{I^2} = \frac{4.4}{4\times10^{-6}}\mu = 1.1\times10^6\,\Omega$

$P' = \frac{V^2}{R} = \frac{(11)^2}{1.1\times10^6} = 11\times10^{-5}$ W

125. (b) When resistances are connected in parallel.

$P_r = \frac{P_1P_2}{P_1+P_2} = 60$

$\because P1 = P2$

$\therefore P1 = P2 = 120$ W

So total consumed power

$P' = P1 + P2 = 120 + 120 = 240$ W

126. (b) $Q = P\times t$

$Q = mc\Delta T + mL$

$P = \frac{V^2_{rms}}{R}$

$4200\times80 + 2260\times10^3 = \frac{(200)^2}{20}\times t$

$\Rightarrow t = 1298$ sec $= 21.6$ min

$t \approx 22$ min

127. (a) By colour coding,

$R = 50\times10^2\,\Omega$

$I^2R = P$

The maximum current which can be passed through the resistor

$I = \sqrt{\frac{P}{R}} = 20mA$

128. [5] The equivalent resistance

$R_{eq} = R_1 + \frac{R_2R_{shunt}}{R_2+R_{shunt}}$,

where $R_1 = 150.4\Omega$, $R_2 = 240\Omega$, and $R_{shunt} = 10\Omega$. Thus,

$R_{eq} = 150.4 + \frac{240\times10}{240+10} = 150.4 + 9.6 = 160\Omega$

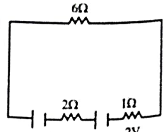
The total current $I_1 = \frac{V}{R_{eq}} = \frac{20}{160} = 0.125$ A

The ammeter current

$I_2 = I_1\times\frac{9.6}{240} = 0.125\times\frac{9.6}{240}$.

$= 0.005$ A $= 5$ mA

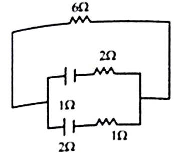
129. [4]



$\varepsilon_{eq} = 3$

$R_{eq} = 9\Omega$

$i = \frac{3}{9} = \frac{1}{3}$



$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1}+\frac{\varepsilon_2}{r_2}}{\frac{1}{r_1}+\frac{1}{r_2}}$

$r_{eq} = \frac{2\times1}{3} + 6 = \frac{20}{3}\Omega$

$i_1 = \frac{1}{4} \Rightarrow \frac{i_1}{i_2} = \frac{4}{3}$

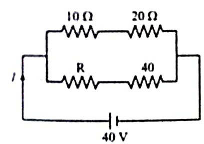
130. (d) The equivalent EMF of two non-ideal batteries connected in parallel is given by:

$E_{eq} = \frac{E_1r_2 + E_2r_1}{r_1+r_2}$

where E_1, E_2 are the EMFs of the two batteries and r_1, r_2 are their internal resistances.

131. [2] This balanced is wheatstone bridge.

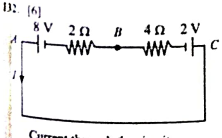
So, $\frac{10}{R} = \frac{20}{40}$



$R = 20\Omega$

$I = \frac{40}{20} = 2A$

132. [6]



Current through the circuit,

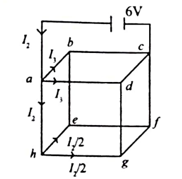
$I = \frac{8-2}{2+4} = \frac{6}{6} = 1$ A

Applying Kirchhoff from C to B

$V_C - 2 - 4\times1 = V_B$

$V_C - V_B = 6V$

133. (1)



From symmetry, current through $e - b$ & $g - d = 0$

$\therefore R_{eq} = \frac{3}{4}\times R = \frac{3}{2}\Omega$

Current through battery $= \frac{6\times2}{3} = 4A$

and $I_2 = \frac{4}{8}\times2 = 1A$

$\therefore \Delta V$ across e–f

$= \frac{I_2}{2}\times R = \frac{1}{2}\times2 = 1V$

134. (a)

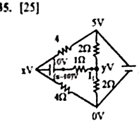
Using ohm's law, $V = IR$

Current in the circuit, $i = \frac{3}{1+2} = 1A$

$v = E - ir$

$= 3 - 1\times1 = 2V$

135. [25]



Applying KCL,

$\frac{y-5}{2} + \frac{y-0}{2} + \frac{y-x+10}{1} = 0$

$y - 5 + y + 2y - 2x + 20 = 0$

$4y - 2x + 15 = 0$ ...(i)

$\frac{x-5}{4} + \frac{x-0}{4} + \frac{x-10-y}{1} = 0$

$x - 5 + x + 4x - 40 - 4y = 0$

$6x - 4y - 45 = 0$ ...(ii)

From equation (i) and (ii), we get

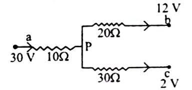
$x = \frac{15}{2}$ & $4y - 15 + 15 = 0$

$y = 0$

$\therefore i = \frac{y-x+10}{1} = \frac{0-7.5+10}{1}$

$\Rightarrow i = 2.5A = \frac{n}{10}A \Rightarrow n = 25$

136. (a) Using kirchhoff's law at point P.



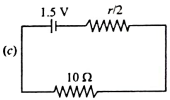
$\frac{P-30}{10} + \frac{P-12}{20} + \frac{P-2}{30} = 0$

$P\left(\frac{1}{10}+\frac{1}{20}+\frac{1}{30}\right) = \frac{30}{10}+\frac{12}{20}+\frac{2}{30}$

$P\left(\frac{6+3+2}{60}\right) = \frac{108+36+4}{60} = 20V$

$\therefore I_{20\Omega} = \frac{x-12}{20} = \frac{20-12}{20} = 0.4A$

137. (c)



$E_{eq} = \frac{\frac{E_1}{r_1}+\frac{E_2}{r_2}}{\frac{1}{r_1}+\frac{1}{r_2}} = \frac{\frac{E}{r}+\frac{E}{r}}{\frac{1}{r}+\frac{1}{r}}$

$= \frac{\frac{2E}{r}}{\frac{2}{r}} = \frac{\frac{2E}{r}}{\frac{2}{r}} = E$

$r_{eq} = \frac{r_1r_2}{r_1+r_2} = \frac{rr}{2r} = \frac{r}{2}$

$Eeq = E = 1.5$ V

$r_{eq} = \frac{r}{2}$

$\therefore V = E - \frac{ir}{2}\quad 1.2 = 1.5 - \frac{ir}{2}\quad \frac{ir}{2} = 0.3$

Again,

$E_{eq} - i'\,10 - \frac{ir}{2} = 0$

$\frac{1.5}{10+\frac{r}{2}} = i$

$10i + \frac{ir}{2} = 1.5$

$10i + 0.3 = 1.5 \Rightarrow i = 0.12$ A

$\therefore r = \frac{0.6}{0.12} = 5\,\Omega$

138. (a) We have,

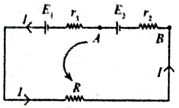
$E_1 = E_2 = E$

$\therefore I = \frac{V_{net}}{R_{net}} = \frac{2E}{R + r_1 + r_2}$

Potential drop across second cell is

$V_A - E_2 + Ir_2 = V_B$

$V_A - V_B = E_2 - Ir_2 = 2$



According to question $V_A - V_B = 0$

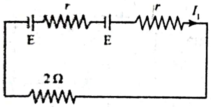
$E_2 - Ir_2 = 0$

$\Rightarrow E - \left(\frac{2E}{R + r_1 + r_2}\right)r_2 = 0$

$\Rightarrow R + r_1 - 2r_2 = 0$

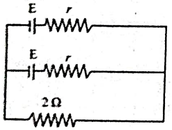
$\Rightarrow R = r_2 - r_1$

139. (a) **Case I:** when the cells are connected in series.



$I_1 = \frac{2E}{2r + 2}$ ...(i)

Case II: when the cells are connected in parallel



$I_2 = \frac{E}{\frac{r}{2} + 2} = \frac{2E}{r + 4}$ ...(ii)

According to the question, $I1 = I2$

From (i) and (ii),

$\frac{2E}{2r+2} = \frac{2E}{r+4} \Rightarrow \frac{2}{2r+2} = \frac{2}{r+4}$

$\Rightarrow 2r + 2 = r + 4$

$\Rightarrow r = 2\,\Omega$

140. (d) Equivalent resistance of the circuit.

$R_{eq} = 5 + \frac{10 \times 5}{10 + 5} + 10 = \frac{55}{3}\,k\Omega$

From Ohm's law,

$V_{AB} = R_{eq} I$

$= \left(\frac{55}{3} \times 10^3\right) \times (15 \times 10^{-3}) = 275\ V$

141. (d) $r_{eq} = r_1 + r_2 + R$

$\therefore I = \frac{2E}{r_1 + r_2 + R}$

According to the question,

$-(V_A - V_B) = E - Ir_2$

$0 = E - Ir_2$

$E = Ir_2$

$E = \frac{2E}{r_1 + r_2 + R} r_2$

$r_1 + r_2 + R = 2r_2$

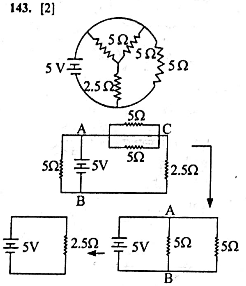
$\therefore R = r_2 - r_1$

142. [4] From Nodal analysis,

$\frac{V_0 - 2}{1k\Omega} + \frac{V_0 - 4}{1k\Omega} + \frac{V_0 - 6}{1k\Omega} = 0$

$V_0 = 4\ V$

143. [2]



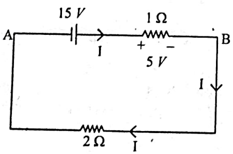
Current supplied by 5V battery

$= 5\ V / 2.5\Omega = 2\ A$

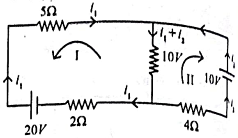
144. [10] $I = \frac{15}{3} = 5A$

$\therefore$ VAB $= 15V - 5 \times 1\ V$

$= 15V - 5V = 10\ V$



145. (c)



Loop-I:

$-20 + 2I_1 + 10(I_1 + I_2) + 5I_1 = 0$

$17I_1 + 10I_2 = 20$ ...(i)

Loop-II:

$-10 + 4I_2 + 10(I_1 + I_2) = 0$

$10I_1 + 14I_2 = 10$ ...(ii)

Equation (i) × (10) – Equation (ii) × 17

$170I1 + 100I2 = 200$

$-170I_1 + 238I_2 = 170$

$-138I_2 = 30$

$I_2 = -\frac{30}{138} = -0.217A$

"–ve" sign indicates that current flows from "+ve" to "–ve" terminal in 10 V battery.

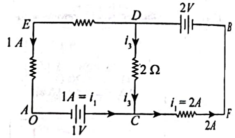
146. (a) $\varepsilon - i_1 R - i_1 R = 0$

$8 - 4i_1 - 4i_1 = 0,\ i_1 = 1A$

147. (a) Let the potential at $A = V_A = 0$

Now at junction C, according to KCL, net current flow at point C is zero.

$i_1 + i_3 = i_2 \quad \Rightarrow i3 = i2 - i1 = 2 - 1 = 1A$

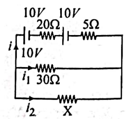


Now analyse potential along $ACDB$;

$V_A + 1 + i_3(2) - 2 = V_B$

$0 + 1 + 2(1) - 2 = V_B,\ V_B = 3 - 2 = 1V$

148. [30]



$V_1 = \varepsilon_1 - i.r_1$

$0 = 10 - i \times 20 \Rightarrow i = 0.5\ A$

$V_2 = \varepsilon_2 - ir_2$

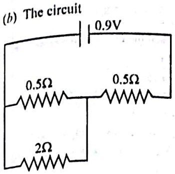
$= 10 - 0.5 \times 5 \Rightarrow V_2 = 7.5\ V$

$0.5 = \frac{7.5}{30} + \frac{7.5}{x}$

$0.5 = 0.25 + \frac{7.5}{x}$

$\frac{7.5}{x} = 0,25;\ x = \frac{7.5}{0.25} = 30$

149. (b) The circuit



$\therefore R_{eq} = 0.5 + \frac{0.5 \times 2}{2 + 0.5} = \left(\frac{5}{10} + \frac{10}{25}\right)\Omega$

$= \frac{45}{50} = 0.9$

$\therefore i = \frac{0.9}{0.9} = 1\ A$

150. (d) $S = \frac{i_g G}{i - i_g}$

$S = \frac{20 \times 10^{-3} \times 30}{2.98}$

$S = \frac{20 \times 30}{2980}$

$\boxed{S = \frac{30}{149}\Omega}$

151. (c) For null point, $\frac{4.5}{60} = \frac{R}{40}$

As Resistance, $R = \frac{\rho\ell}{A} = \frac{\rho\ell}{\pi r^2}$

$153...5 \times 40 = \rho \times \frac{0.1}{\pi \times 7 \times 10^{-8}} \times 60$

$\Rightarrow \rho = 66 \times 10^{-7}\Omega \times m$

152. (d) As dimension of wire does not effect specific resistance so, it will not change.

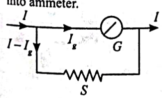
153. (d) According to Kirchoff voltage is low $V_1 + V_2 - V_3 = 0 \Rightarrow V_1 + V_2 = V_3$

154. (c) Given $G = 10\,\Omega$

$I_g = 3$ mA

$I = 8$A

In case of conversion of galvanometer into ammeter.



We have $I_g G = (I - I_g)S$

$S = \frac{I_g G}{I - I_g}$

$S = \frac{(3 \times 10^{-3})10}{8 - 0.003} = 3.75 \times 10^{-3}\,\Omega$

155. (c) Since all the resistance in series, Hence, $R_{eq} = 4000\,\Omega$

So current passing through circuit is,

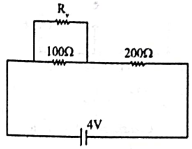
$I = \frac{4}{4000} = \frac{1}{1000}A$

Hence, $V_0 = I.R = \frac{1}{1000} \times 500 = 0.5\ V$

156. (200) Given $R_1 = 100$ ohm

$R_2 = 200$ ohm

$V = 4$ V



$\frac{R_v 100}{R_v + 100} = \frac{200}{3}$

$3R_v = 2R_v + 200$

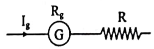
$R_v = 200$

157. (c) $i = \frac{E_{eq}}{r_{eq}} = \frac{8 \times 5}{8 \times 0.2} = 25A$

As, $V = E - ir$

$= 5 - 0.2 \times 25 = 0$

158. (c) Let the resistor to be connected in series with R_G be R.

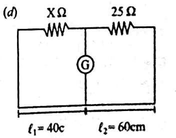


$R = \frac{V}{I_g} - R_g = \frac{100}{5 \times 10^{-3}} - 50$

$= 20000 - 50$

$= 19950\Omega$

159. (d)



Balance equation is given as,

$\frac{25}{r\ell_1} = \frac{X}{r\ell_2}$ ...(i)

$\frac{25}{2r\ell_1} = \frac{X}{2r\ell}$ ...(ii)

Solving (i) and (ii), we get

$\ell'_1 = \ell_1 = 40$ cm

160. (Bonus) By using,

$I_g S = I_s G$

$\Rightarrow \frac{95}{100} IS = \frac{5I}{100} G$

$\Rightarrow S = \frac{G}{19}$

$R_s = \frac{SG}{S + G} = \frac{\frac{G^2}{19}}{\frac{20G}{19}}$

$\therefore R_s = \frac{G}{20}$

161. (c) For zero deflection $\frac{R}{3} = \frac{0.8}{1}$

$\Rightarrow R = 2.4\ m\Omega$

It is given that temperature fall in 10s $= 20°C$

$\Delta R = R\,\alpha\,\Delta t$

$\alpha = \frac{\Delta R}{R\Delta t} = \frac{-0.6}{3 \times 20}$

$= -10^{-2}C^{-1}$

162. (a) So, First case $\frac{2}{40} = \frac{X}{60} \Rightarrow X = 3$

In second case $X' = \boxed{\frac{2 \times 3}{2 + 3}} = 1.2\Omega$

$\frac{2}{\ell} = \frac{1.2}{100 - \ell}$

$200 - 2\lambda = 1.2\lambda$

$\ell = \frac{200}{3.2} = 62.5$ cm

$\therefore$ Shift in balance point $= 22.5$

163. (4) As $V = IR = E - Ir$

$\therefore i_1 \times 10 = \lambda \times 500 = \varepsilon - i_1$

[Let potential gradient be λ.]

$\Rightarrow 500\lambda = \varepsilon - 50\lambda r_e$ ...(1)

Also, $i_2 \times 1 = \lambda \times 400 = \varepsilon - i_2$

$\Rightarrow 400\lambda = \varepsilon - 400\lambda r_e$ ...(2)

From eq^n (1) & eq^n (2)

$\therefore 100\lambda = 350\lambda r_e \Rightarrow r_e = \frac{10}{35} = 0.3\Omega$

164. (c) The current in the galvanometer,

$i_g = \frac{10}{400 + 100} = 20 \times 10^{-3}\,A$

For ammeter

Let shunt resistance $= r$

$i_g R = (i - ig)r$

$20 \times 10^{-3} \times 100 = 10\,r$

$r = 20 \times 10^{-2}\,\Omega$

165. (2500)

The equivalent resistance

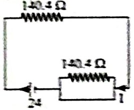
$$R_{eq} = \frac{10^4 R}{10^4 + R}$$

$V = 4V, I = 2mA$

$$I = \frac{V}{Req} \Rightarrow 2\times10^{-3} = \frac{4(10^4 + R)}{10^4 R}$$

$\Rightarrow 20R = 40000 + 4R \Rightarrow R = 2500\Omega$

166. (160)



$$R_{eq} = 140.4 + \frac{240\times10}{240+10}$$

$R_{eq} = 150\ \Omega$

Reading of ammeter $= \frac{24}{150}$

$= 160$ mA

167. (b) Given:

$G = 200\ \Omega$ & $i_g = 20\ \mu A$

$$i = i_g\left(\frac{G}{S}+1\right)$$

$$\Rightarrow 20\times10^{-3} = 20\times10^{-6}\left(\frac{200}{S}+1\right)$$

$\Rightarrow S \approx 0.2\ \Omega$

168. (c) For voltmeter, Apply Kirchhoff's law $V-(I_g \cdot G)$

$-I_g \cdot R = 0$

$$R = \frac{V}{I_g} - G$$

$$= \frac{50}{10^{-3}} - 54 = 49946$$

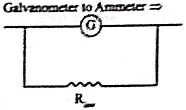
$= 49.946k\Omega = 50k\Omega$

For ammeter, $Ig \cdot G - (I - Ig)r = 0$

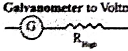
$$r = \frac{I_g G}{I - I_g} = \frac{10^{-3}\times 54}{(10-1)\times10^{-3}} = 6\Omega$$

169. (a) Error of voltmeter decreases with increase in its resistance.

170. (c) Galvanometer to Ammeter ⇒



Galvanometer to Voltmeter ⇒



171. [30]

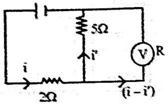
$$\frac{R_1+R_2}{10} = \frac{60}{40} \Rightarrow R1 + R2 = 15$$

Now $\frac{R_1R_2}{(R_1+R_2)\times3} = \frac{40}{60}$

$\Rightarrow R_1R_2 = 30\Omega^2$

172. [20]

$$i' = \frac{2V}{5\Omega} = \frac{2}{5}A$$



$$i = \frac{(3-2)}{2} = \frac{1}{2}A$$

∴ Current through voltmeter

$$= \frac{1}{2} - \frac{2}{5} = \frac{1}{10}A$$

∴ For voltmeter $V = (i - i')R$

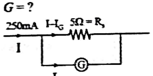
$$2 = \left(\frac{1}{10}\right)R \Rightarrow R = 20\Omega$$

173. [50]

Given, $I = 250$ mA $= 0.250$ A

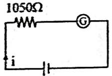
$R_s = 5\Omega$

$G = ?$



$(I - IG)R_s = IG \cdot G$

$$\Rightarrow I_G = \frac{IRs}{Rs+G} = \frac{0.250\times5}{5+G} \quad ...(i)$$



$$I_G = \frac{25}{1050+G} \quad ...(ii)$$

From (i) and (ii)

$$\frac{0.250\times5}{5+G} = \frac{25}{1050+G}$$

$G = 50\Omega$

174. [2] For a balanced meter – bridge,

$$\frac{R}{S} = \frac{l}{100-l}$$

$$\Rightarrow \frac{2}{\left(\frac{3X}{3+X}\right)} = \frac{40+22.5}{60-22.5} = \frac{5}{3}$$

$$\Rightarrow \frac{6}{5} = \frac{3X}{3+X}$$

$\Rightarrow x = 2$

175. [5] When the external resistance, $R = 5\Omega$

$$\frac{\varepsilon}{r+5}\times5 = 200x \quad ...(i)$$

When the external resistance, $R = 15\Omega$

$$\frac{\varepsilon\times15}{r+15} = 300x \quad ...(ii)$$

Dividing eqn (i) by (ii)

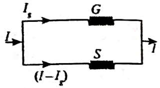
$\Rightarrow r = 5\Omega$

176. (b) Given, $G = 72\Omega$, $S = 8\Omega$

$\Rightarrow Ig\ G = (I - Ig)S$

$$\Rightarrow \frac{I - I_g}{I_g} = \frac{G}{S}$$

$$\Rightarrow \frac{I}{I_g} - 1 = \frac{72}{8}$$



$$\Rightarrow \frac{I}{I_g} = 10$$

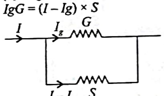
$$\Rightarrow \frac{I_g}{I} = \frac{1}{10}$$

$$\Rightarrow \frac{I_g}{I}\times100 = 10\%$$

% of current which passes through galvanometer is 10%.

177. (b) In galvanometer

$IgG = (I - Ig) \times S$



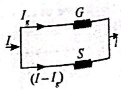
$$\frac{I_g}{I} = \frac{S}{S+G} \Rightarrow \frac{1}{3} = \frac{S}{S+G} \Rightarrow G = 2S$$

178. (b) Given, $G = 72\Omega$, $S = 8\Omega$

$\Rightarrow Ig\ G = (I - Ig)S$

$$\Rightarrow \frac{I - I_g}{I_g} = \frac{G}{S}$$

$$\Rightarrow \frac{I}{I_g} - 1 = \frac{72}{8}$$



$$\Rightarrow \frac{I}{I_g} = 10$$

$$\Rightarrow \frac{I_g}{I} = \frac{1}{10}$$

$$\Rightarrow \frac{I_g}{I}\times100 = 10\%$$

% of current which passes through galvanometer is 10%.

179. (d)

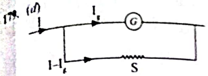


Figure of merit $\frac{I_g}{\theta} = K$

$I_g = Kn$

$$I = \frac{I_g}{s}(G+S)$$

$$\therefore I = \frac{nK}{S}(G+S)$$

180. [2400] $\frac{S}{R} = \frac{l}{100-l} \Rightarrow S = 5.6\times\frac{30}{100-30}$

$$\Rightarrow S = \frac{3}{7}\times5.6k\Omega = 3\times0.8k\Omega$$

$= 2.4k\Omega = 2400\Omega$

181. [19]

Unknown resistance R

Known resistance, $P = 15\Omega$

At null point

$$\frac{P}{R} = \frac{l}{100-l}, \frac{15}{R} = \frac{43+2}{100-(43)}$$

$R = 19\Omega$

182. [40]

Since, potential gradient is independent of area therefore, balancing point will be same .i.e.

$x = 40$ cm

183. [10]

Using the balance condition of Wheatstone bridge:

$$\frac{R}{3} = \frac{4}{6} \Rightarrow R = 2\Omega$$

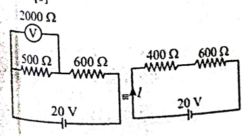
Equivalent resistance of the circuit $Req =$

$$\frac{9\times6}{9+6} = \frac{54}{15}\Omega$$

From ohm's law,

$$i = \frac{36}{\frac{54}{15}} = 10A$$

184. [8]



$$I = \frac{20}{400+600} = 20 \text{ mA}$$

Current through the voltmeter =

$$I\times\frac{500}{2500} = 20\times\frac{500}{2500} = 4 \text{ mA}$$

∴ Voltage across the voltmeter = Current through in voltmeter × voltmeter resistance

$= 4 \times 10\text{-}3 \times 2000 = 8$ V

185. [20]

Initially, $\frac{P}{Q} = \frac{40}{60} = \frac{2}{3}$ ...(i)

$$\frac{4}{Q} = \frac{2}{3} \Rightarrow Q = 6\Omega$$

Finally, $\frac{P+x}{Q} = \frac{80}{20} = \frac{4}{1}$ ...(ii)

$$\frac{4+x}{6} = \frac{4}{1}$$

$x = 24 - 4 = 20\Omega$

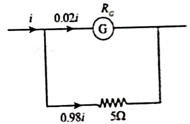
186. (a) Current at full deflection,

$$I_{max} = \frac{\text{Total number of division}}{\text{Sensitivity}}$$

$$= \frac{50}{2} = 25\text{mA}$$

$$R = \frac{V}{I_{max}} = \frac{50}{25} = 2\Omega$$

187. (d)



For Ammeter

$I_G R_G = (I - I_G)s$

$$\frac{2i}{100}R_G = \frac{98i}{100}\times5$$

$R_G = 245\Omega$

188. (c) Let, $V_A = 10\ V$, $VB = x\ V$, $V_c = 0\ V$, and $V_D = y\ V$

Net current from B is zero, so,

$$\frac{x-10}{100} + \frac{x-y}{15} + \frac{x-0}{10} \quad ...(i)$$

$= 0 \Rightarrow 53x - 20y = 30$

Net current from D is zero, so,

$$\frac{y-10}{60} + \frac{y-x}{15} + \frac{y-0}{5} \quad ...(ii)$$

$= 0 \Rightarrow 17y - 4x = 10$

Solving equations (i) and (ii), we get,

$x = 0.865$ and $y = 0.792$

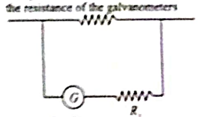
$$i_G = \frac{x-y}{15} = 4.87 \text{ mA}$$

189. [48]

Using the galvanometer coil balanced formula, under balanced condition,

$$\frac{12}{x} = \frac{6}{72-x} \Rightarrow x = \frac{144}{3}\ 48 \text{ cm}$$

190. (a) I_g is current in the galvanometers G is the resistance of the galvanometers



$V = I_g(G + R_1)$

$1 = I_g(G + R_1)$

$2 = I_g(G + R_1 + R_2)$

$2 = 1 + I_g R_2$

$I_g R_2 = 1$

$R_2 = G + R_1$

191. (c) For parallel combination of 10 kΩ and 400 Ω

$$R = \frac{40000}{104} = 384.61\Omega$$

The potential difference across 400 Ω is

$$V400\ \Omega = \frac{384.61}{384.61+300}\times6 = 1.95\ V$$

192. (d) Current passing through the galvanometer

$I = 6$mA

Deflection in galvanometers, $\theta = 2°$

Figure of Merit

$$= C = \frac{i}{\theta} = \frac{6\times10^{-3}}{2} = 3\times10^{-3}\ Am^2$$

193. (a) Voltage in the galvanometer,

$V_g = i_g R_g = 10^{-3} \times 100$

$$= \frac{100}{1000} = \frac{1}{10}$$

Potential difference, $V = 10$ V

$$= R_g\left(\frac{V}{V_g} - 1\right)$$

$= 100 \times 99 = 9.9$ KΩ

194. [40] $\frac{X}{R} = \frac{75}{25} = 3$, $R = \frac{4\rho\ell}{\pi d^2}$ ...(i)

$$R' = \frac{4\rho\left(\frac{\ell}{2}\right)}{\pi\left(\frac{d}{2}\right)^2} = 2R \quad ...(ii)$$

then $\frac{X}{R'} = \left(\frac{100-\ell}{\ell}\right)$

$$\frac{100-\ell}{\ell} = \frac{X}{2R} = \frac{3}{2}$$

$\ell = 40.00$ cm

195. [2] Current across 90Ω and 60Ω resistance

$i_1 = \frac{40}{100} = \frac{2}{5}$

Current across 90Ω and 110Ω resistance

$i_2 = \frac{40}{200} = \frac{1}{5}$

$V_A - V_B = 40i_1 = 40\times\frac{2}{5} = 16$ ... (i)

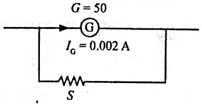
$V_A - V_D = 90I_2 = \frac{90}{5} = 18$ ... (ii)

From (i) and (ii) we get

$V_B - V_D = 18 - 16 = 2$ volt

Hence, potential difference across BD is 2 volt

196. (a) $S = \frac{50\times0.002}{(0.5-0.002)}$ $S = 0.2\Omega$



197. (b) G = 50Ω, S = 5000 Ω, $Ig = 4\times10^{-3}$

Max. voltage, V = Ig(G + S)

V = 4 × 10–5 (50 + 5000) = 20.2 volt.

198. (c) Maximum current through galvanometer, I_{max}
$= 50\times20\times10^{-6} = 10^{-3}$ A

(i) $I_{max} = \frac{2}{100+R_1} = 10^{-3} \Rightarrow R_1 = 1900\Omega$

(ii) $I_{max} = \frac{10}{R_G+R_1+R_2}$

$10^{-3} = \frac{10}{100+1900+R_2} \Rightarrow R_2 = 8000\Omega$

(iii) $I_{max} = \frac{20}{100+1900+8000+R_3}$

$R_3 = 10000\Omega$

199. (a) We know that, $x = \frac{R(100-l)}{l}$

for (1) $x = \frac{1000\times(100-60)}{60} \approx 667$

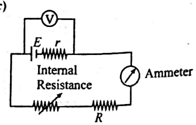
for (2) $x = \frac{100\times(100-13)}{13} \approx 669$

for (3) $x = \frac{10\times(100-1.5)}{1.5} \approx 656$

for (4) $x = \frac{1\times(100-1)}{1} \approx 99$

So reading in serial no. (4) is completely different hence correct answer (a).

200. (c)



$V = E - Ir$

When $V = V_0 = 0 \Rightarrow 0 = E - Ir$

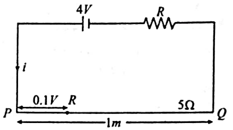
$\therefore E = Ir$

When $I = 0$, $V = E = 1.5$ V

$\therefore r = 1.5\ \Omega$

201. (c) Current, $i = \frac{4}{5+R}$

$V_{PQ} = i(5) = \frac{20}{5+R}$



$V_{PR} = \frac{V_{PQ}}{L}(0.1) = \frac{20}{5+R}\left(\frac{0.1}{1}\right) = \frac{2}{5+R}$

Now, $\frac{2}{5+R} = 5\times10^{-3}$

2 = 25 × 10–3 + 5 × 10–3 R

$R = \frac{2-0.25}{5\times10^{-3}} = 395\ \Omega$

202. (*) According to ohm's law V = IR

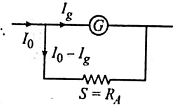
⇒ V = Ig = (R + G)

⇒ 5 = 10–4 × (2 × 10^4 + x)

⇒ x = –195 × 10^4 Ω (Bonus)

203. (b) When galvanometer is used as an ammeter, shunt is used in parallel with galvanometer.

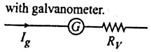
∴



$\therefore I_g G = (I_0 - I_g)R_A$

$\therefore R_A = \left(\frac{I_g}{I_0 - I_g}\right)G$

When galvanometer is used as a voltmeter, resistance is used in series with galvanometer.



$I_g(G + R_V) = V = GI_0$ (given $V = GI_0$)

$\therefore R_V = \frac{(I_0 - I_g)G}{I_g}$

$\therefore R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

204. (c) $\frac{R_1}{R_2} = \frac{2}{3}$ ...(i)

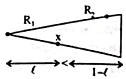
and $\frac{R_1+10}{R_2} = 1$ ...(ii)

After solving (i) and (ii)

R2 = 30Ω and R1 = 20Ω

Now, $R1 = \frac{30\times R}{30+R} = 20 \Rightarrow R = 60\Omega$

205. (c) $\frac{R'}{R_1} = \frac{R'}{R_2} \Rightarrow R_1 = R_2 = \frac{R}{2}$



Total resistance = R

$= \int_{\text{wire}} dR = \int_0^1 \frac{k}{\sqrt{l}}\cdot dl = k\left[2\sqrt{l}\right]_0^1 = 2k$

$R_1 = \frac{R}{2} = k \Rightarrow \int_0^{Q} \frac{k}{\sqrt{l}}\cdot dl = k$

$[2k\sqrt{l}]_0^l = k,\ \sqrt{l} = \frac{1}{2}, l = 0.25$ m

206. (b) $i_g = 4\times10^{-4}\times25 = 10^{-2}$ A

$V_o = i_g(R_G + R)$

$\therefore R_g + R = \frac{V_o}{i_g} = \frac{2.5}{10^{-2}} = 2.5\times10^2 = 250$

⇒ R = 250 − 50 = 200 Ω

207. (d) Let the length AJ, be x when galvanometer shows no deflection

$V_{AJ} = IR_{AJ}\frac{E}{3} = \left(\frac{E}{12r+3r}\right)\times\left(\frac{x}{L}\times12r\right)$

$x = \frac{15r\times L}{12r\times3} = \frac{5}{12}L$

208. (c) Rg = 20Ω

Division on both side = 30

FOM = $\frac{1}{\phi}$ = 0.005 A/Div.

Current sensitivity = $\frac{1}{FOM}$ = 0.005 Div/A

Maximum current = 0.005 × 30 = 0.15

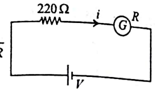
V = Ig(max)(20 + R)

15 = 0.15[20 + R], 100 = 20 + R

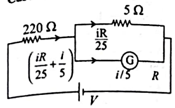
⇒ R = 80Ω

209. (b) **Case 1:** K1 and K2 are open.

$i = \frac{V}{220+R}$



Case 2: K1 and K2 are closed.



$\left(\frac{iR}{25}+\frac{i}{5}\right)220 + \frac{i}{5}R = V$

$\left(\frac{iR}{25}+\frac{i}{5}\right)220 + \frac{i}{5}R = i(220+R)$

After solving R = 22 W

210. (d) We know that, $\frac{P}{Q} = \frac{R}{S}$

Case 1: $\frac{P}{Q} = \frac{400}{x}$, Case 2: $\frac{Q}{P} = \frac{405}{x}$

Using case 1 and case 2:

x2 = 400 × 405, x = 402.5 Ω

211. (b) When Rh = 2 Ω

$i_1 = \frac{6}{4+2} = 1A$

$E_1 = \frac{1\times4}{AB}\times AJ \Rightarrow AJ = \frac{AB}{4}E_1$

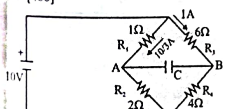
When Rh = 6 Ω

$i_2 = \frac{6}{4+6} = 0.6A$

$E_2 = \frac{0.6\times4}{AB}\times AJ \Rightarrow AJ = \frac{AB}{2.4}E_2$

∴ E2 = 0.6 × E1 = 0.6 × 0.5 = 0.3 V

212. [400]



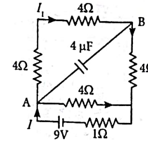
$V_A + \frac{10}{3}(1) - 6(1) = V_B$

$\Rightarrow V_A - V_B = 6 - \frac{10}{3} = \frac{8}{3}$ volt

Accumulated charge on C (Q) = C($V_A - V_B$)

$= 150\times\frac{8}{3} = 400\mu C$

213. [81]



$I = \frac{V}{R_{eq}} I = \frac{V}{R_{eq}} = \frac{9}{1+\frac{12\times4}{12+4}} = \frac{9}{4}$

$I_1 = \frac{9}{4}\times\frac{4}{16} = \frac{9}{16}$

$V_A - V_B - V_1\times8 = \frac{9}{16}\times8 = \frac{9}{2}$ V

The energy stored by the capacitor will be

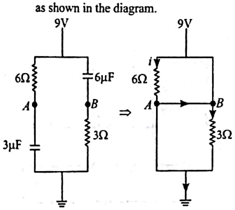
$\therefore U = \frac{1}{2}\times4\times\frac{81}{4}\mu J$

$\therefore U = \frac{81}{2}\mu J$

$\therefore x = 81$

214. [36]

At steady state, capacitor behaves as an open circuit and current flows in circuit as shown in the diagram.



Equivalent resistance,

$R_{eq} = 9\Omega$

$i = \frac{9V}{9\Omega} = 1A$

$\Delta V_{6\Omega} = 1\times6 = 6V$

$V_A = 3V$

So, potential difference across 6μF is 6V.

The charge stored in 6(mu)F capacitor,

Hence, $Q = C\Delta V$

$= 6\times6\times10^{-6}$ C = 36 μC

215. (d) In steady state

Req = 12Ω

$I = \frac{6}{12} = 0.5A$

Now, $q_1 = C_1V_1 = 12\ \mu C$

$q_2 = C_2V_2 = 24\ \mu C$

$\because \frac{q_1}{q_2} = \frac{1}{2}$

216. [60]

No voltage drop across $R_2 = 5\Omega$, because in steady state there will be no current in branch of capacitor

$\Rightarrow I_2 = 0$

$I_1 = I_3 = \frac{10}{4+6} = 1A$

$V_{R_3} = V_c + V_{R_2}$ & $V_{R_2} = 0$

$I_3R_3 = V_c$

$V_c = 1\times6 = 6$ volt

$q_c = CV_c = 10\times6 = 60\ \mu C$

217. [4] Let initial electric field between plates be E_0,

$E = \frac{E_0}{3} \Rightarrow V = \frac{V_0}{3}$

$\frac{V_0}{3} = V_0 e^{-\frac{t}{\tau}}$

$t = \tau \ln 3$

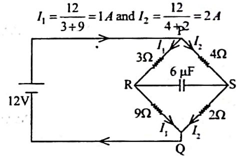
$6.6\times10^{-6} = R(1.5\times10^{-6})(1.1)$

$R = \frac{6}{1.5} = 4\Omega$

218. [75]

From ohm's law

$I_1 = \frac{12}{3+9} = 1A$ and $I_2 = \frac{12}{4+2} = 2A$



$V_P - V_R = 3I_1 = 3V$ ... (i)

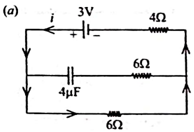
$V_P - V_S = 2\times4 = 8V$ ... (ii)

Subtracting eq. (i) from eq. (ii)

$V_R - V_s = 5V \Rightarrow V = 5V$

$U = \frac{1}{2}CV^2 = \frac{1}{2}\times6\times5^2 = 75\mu J$

219. (a)



In steady state, current do not pass through capacitor, therefore $I = \frac{3}{6+4} = \frac{3}{10}$

Potential difference on 6Ω resistance

$V = 6 \times \frac{3}{10} = 1.8$ volt

Capacitor will have same potential difference there fore, charge on it $q = CV$ $= (4\ \mu F).(1.8\ \text{volt}) = 7.2\ \mu C$.

220. (b) At steady state, capacitors are open.

$\therefore i = \frac{4V}{6+2+8} = \frac{1}{4}A$

Voltage across C_1 is

$V_1 = V_{AC} = i(6\Omega + 2\Omega) = 8i$ Volt

Voltage across C_2 is

$V_2 = V_{BD} = i(2\Omega + 8\Omega) = 10i$ Volt

$\Rightarrow \frac{V_1}{V_2} = \frac{8i}{10i} = \frac{4}{5}$

221. (*d*) $q = q_0\, e^{-t/Rc}$

(discharging eq. of capacitor)

$t \to \frac{t}{2} \Rightarrow q \to \frac{q_0}{\sqrt{2}}$

$\frac{q_0}{\sqrt{2}} = q_0 e^{-t_1/\tau}$

$\sqrt{2} = e^{-t_1/\tau} \Rightarrow t_1 = \tau \ell n \sqrt{2}$

$\frac{q_0}{8} = q_0 e^{-t_2/Rc}$

$8 = e^{t_2/Rc} \Rightarrow t_2 = \tau \ell n 8$

$\frac{t_1}{t_2} = \frac{\ln\sqrt{2}}{\ln 8} = \frac{\frac{1}{2}\ln 2}{3\ln 2} = \frac{1}{6}$

222. [10]

$q = CV$

$= \left(\frac{11}{10} \times 10^{-6}\right)\left(\frac{100}{11}\right)$

$= 10\ \mu C$

223. (*b*) $V = V0(1 - e{-}t/RC)$

$V = 10(1 - e{-}t/RC)$

$2 = 20(1 - e{-}t/RC)$

$\frac{1}{10} = 1 - e^{-t/RC}$

$e^{t/RC} = \frac{10}{9}$

$\frac{t}{RC} = \ln\left(\frac{10}{9}\right) = 0.105$

$C = \frac{10^{-6}}{10 \times 0.105} = 0.95 \mu F$

224. [100]

We know that, charge on the capacitor, $Q = CV$

$Q = 50\ \mu F \times 2V = 100 \times 10{-}6C$

$Q = 100\ \mu C$

Hence, charge on the upper plate of capacitor is 100 μC

225. (d) Given, $C = 1\ \mu F = 1 \times 10{-}6\ F$

$V0 = 100$ volt, V = 50 volt

R = 100 Ω

Where τ = RC = 100 × 10^{-6} sec

$50V = 100V\left(1 - e^{\frac{-t}{10^{-4}}}\right)$

$\frac{1}{2} = 1 - e^{-10^4 t} \Rightarrow \frac{1}{2} = e^{-10^4 t}$

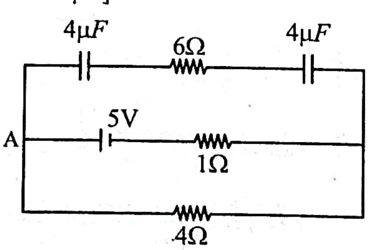
$2 = e{+}104t \Rightarrow \ell n2 = 104t$

$t = \frac{\ell n 2}{10^4}$

$t = 0.693 \times 10^{-4}$ sec.

226. (*d*) $V_{AB} = \frac{5}{5} \times 4 = 4V$

$\Rightarrow Q = Ceq\ V = 2 \times 4 = 8\mu c$ [$\because$ $Ceq = 2 + 2 = 4\mu F$]

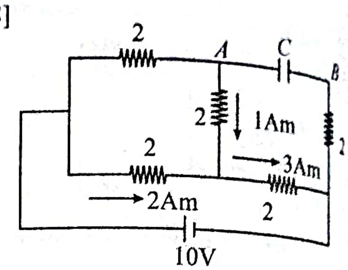


227. [2] $I = \frac{V}{R} = \frac{30/3}{5 \times 10^6} = 2 \times 10^{-6}$

228. (*a*) τ = RC = 10 μs

For $0 < t < 5$ μs, it will get charged. For $5 < t < 10$ μs potential is constant and again gets charged after that.

229. [8]



After solving the circuit we got the final current distribution as shown in the above diagram. So potential difference between A and B is

0 + (2 × 3) + (2 × 1) = 8 volt

230. [6] After connection is common potential

$V_{\text{common}} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

$= \frac{60 \times 20 + 0}{120} = 10V$

Energy loss =

$U_i - U_f = \frac{1}{2}CV^2 - \frac{1}{2}(2C) \times V^2_{\text{Common}}$

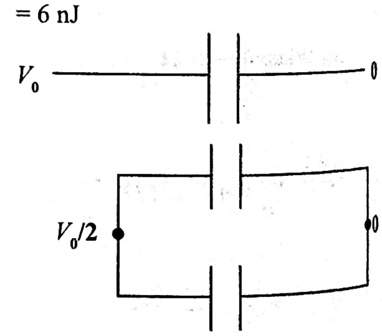
$= \frac{1}{2} \times 60 \times 10^{-12} \times (20)^2$

$- 60 \times 10^{-12} \times (10)^2$

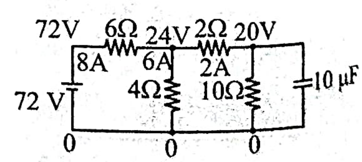
$= 60 \times 10^{-12}\ (200 - 100)$

$= 6000 \times 10^{-12}$

= 6 nJ



231. (*a*) Different potential is shown at different points.



VC = 20 V, C = 10 μF

$\therefore q = CV = 10 \times 10{-}6 \times 20$

$q = 200\ \mu C$